Supplemental Material for: Non-additive Strain–Dopant Energetics in Quasi-Hexagonal C_{60} Graphullerene

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I. FINITE-DIFFERENCE DERIVATION OF THE FOUR-CORNER COUPLING ESTIMATOR

II. DISCRETE STRAIN–SUBSTITUTION INTERACTION

The production estimator is a discrete interaction at fixed dopant species. B, N, and P are chemical labels, not three numerical samples of one continuous coordinate δ. For a fixed species δ and a biaxial load ϵ,

$$\mathcal{S}_\delta(\epsilon) \equiv \frac{E(\epsilon,\delta) - E(\epsilon,0) - E(0,\delta) + E(0,0)}{N_{\text{atoms}}}, \qquad (1)$$

which is the same four-corner combination as Eqs. (2) and (3). It cancels the strain-only and substitution-only increments and retains their joint remainder. It is not $\partial^2 E/\partial\epsilon\partial\delta$.

A mixed derivative appears only after a separate, local composition coordinate x is introduced at that fixed species. In that local representation the cross coefficient below is a finite-difference image of $\partial^2 e/\partial\epsilon\partial x$, not a derivative along a shared B/N/P axis.

A. Local expansion at fixed species

Let $e(\epsilon, x)$ be the total energy per atom, with ϵ the biaxial load and x a local composition coordinate defined only after the species δ has been fixed. About the pristine, unloaded reference $(\epsilon, x) = (0, 0)$,

$$\begin{aligned} e(\epsilon,x) = e_0 + e_\epsilon\,\epsilon + e_x\,x + \tfrac{1}{2}e_{\epsilon\epsilon}\,\epsilon^2 + \tfrac{1}{2}e_{xx}\,x^2 \\ + e_{\epsilon x}\,\epsilon x + \mathcal{O}(3), \end{aligned} \qquad (2)$$

where subscripts denote partial derivatives evaluated at the reference and $e_{\epsilon x} \equiv \partial^2 e/\partial\epsilon\partial x$. The four corners of one species are $e(0,0)$, $e(\epsilon,0)$, $e(0,x)$, and $e(\epsilon,x)$. Forming the inclusion–exclusion combination gives

$$\begin{aligned} e(\epsilon,x) - e(\epsilon,0) - e(0,x) + e(0,0) \\ = e_{\epsilon x}\,\epsilon x + \mathcal{O}(\epsilon^2 x,\ \epsilon x^2,\ \epsilon^3,\ x^3). \end{aligned} \qquad (3)$$

Every term that depends on ϵ alone or on x alone cancels. Dividing by ϵx isolates the mixed derivative of this local chart,

$$\frac{e(\epsilon,x) - e(\epsilon,0) - e(0,x) + e(0,0)}{\epsilon x} = e_{\epsilon x} + \mathcal{O}(\epsilon, x). \qquad (4)$$

The main-text order parameter is this interaction reported per atom at the production load (+3%) and at the discrete label δ, without dividing by a continuous Δx. The same cancellation is what Eq. (3) states for total energies.

B. Why the four-corner quotient is not the local slope

Dividing the per-atom interaction by the load,

$$\frac{\mathcal{S}_\delta(\epsilon)}{\epsilon} = e_{\epsilon x}\,x + \mathcal{O}(\epsilon, x), \qquad (5)$$

still carries the local composition step x. It is therefore not equal to $e_{\epsilon x}$, and it is not a derivative with respect to dopant identity. Equation (4) uses that quotient only as a finite-difference image inside one local chart. The manuscript reports $\mathcal{S}_\delta(\epsilon)$ itself.

C. What the four corners remove

A strain-only scan at fixed $\delta = 0$ retains $e_\epsilon\epsilon + \frac{1}{2}e_{\epsilon\epsilon}\epsilon^2$. A substitution-only scan at fixed $\epsilon = 0$ retains $e_x x + \frac{1}{2}e_{xx}x^2$. Adding those two scans double-counts nothing in the cross term and still omits $e_{\epsilon x}\epsilon x$. The four-corner combination deletes both one-variable polynomials through the order kept in Eq. (2) and retains the mixed term. That is the sense in which sequential scans are additive and the joint evaluation is not: the omitted piece is $e_{\epsilon x}\epsilon x$, up to higher-order remainders.

D. Gauge and higher-order content

Shifting the energy zero cancels inside Eq. (3), so $\mathcal{S}_\delta$ does not depend on a global energy offset. At the production load of +3% and at a full substitution on the doped cell, the steps in ϵ and in local x are not infinitesimal. Cubic and higher terms then sit inside the reported interaction. The manuscript does not separate them from $e_{\epsilon x}\epsilon x$. Table II records that limit in one line: the mixed derivative is defined only inside the local chart, and the higher-order remainder is not resolved.

E. Propagation of corner noise

For four uncorrelated corner energies with equal standard uncertainty σ_E, the numerator of Eq. (1) has un-

TABLE I. Positioning relative to representative prior work on qHP graphullerene and related covalent networks (not an exhaustive literature map). "Joint (ϵ, δ)" denotes simultaneous strain and substitutional doping in the same calculation series; "four-corner total-energy audit" denotes symmetric (ϵ, δ) corner differences as in the main-text definition of $\mathcal{S}$.

Prior focus	Joint (ϵ, δ)	Four-corner E audit	Nonlinear
Separate strain or doping maps [1–7]	×	×	×
Nonlinear band-edge / transport response [8–10]	Partial	×	×
Alloy cluster expansion (effective $c\,\epsilon x$) [11, 12]	✓	×	×
This work (qHP C_{60}, PBE+D3)	✓	✓	✓

certainty $2\sigma_E$. Per atom, with $\sigma_e = \sigma_E/N_{\text{atoms}}$,

$$\sigma_{\mathcal{S}} = 2\sigma_e. \tag{6}$$

This is the same quadrature as Eq. (5). Correlated SCF noise across chemically related corners would reduce the effective coefficient below 2; the main-text reporting band uses $2\sigma_e$ as a conservative reporting floor (Table I). That floor is a statistical envelope on corner noise. It is distinct from the rigid-protocol estimate, which compares the fixed-coordinate evaluation with the fully relaxed fixed-cell evaluation studied in the main text and is not a bound on every constraint manifold.

III. LITERATURE POSITIONING AND DESCRIPTOR COMPARISON

IV. ARCHIVED LEGACY-PBE TETRAMER BASELINE

Table III reproduces the archived tetramer cluster map (PBE without D3; Cartesian coordinates not affinely scaled) retained for reproducibility; it is *not* used for quantitative strain-physics claims in the main text.

V. CALCULATION INVENTORY (REPRODUCIBILITY)

Tables IV and V list the DFT groups underlying the main-text claims and superseded exploratory grids retained only for reproducibility. They are *not* scientific results tables.

VI. FULL TETRAMER π-DOS AT ϵ=0

VII. PERIODIC FRONTIER-ORBITAL LOCALIZATION

Cage-resolved VBM/CBM isosurfaces on periodic $n \times C_{60}$ cells (n = 2–8) at ϵ=0 complement the tetramer π-DOS above and the main-text electronic taxonomy (donor/acceptor vs. size dilution). Carbon cages are shown as uniform gray MSMS molecular surfaces (probe 1.3 Å); heteroatoms use element colors.

VIII. PERIODIC COUPLING-ENERGY ROWS ($n \geq 6$)

IX. FOUR-CORNER TOTAL ENERGIES (+3%)

X. MARCUS REORGANIZATION ENERGIES

Polaron transport diagnostics are reported here for completeness; the main-text analysis centers on $\mathcal{S}$.

XI. HOPPING INTEGRAL J

On the tetramer dimer grid, B substitution at ϵ=0 delocalizes the charged-state orbital (IPR 75.0 → 24.6) while raising J from 26.6 to 46.4 meV; coupled B @ +3% gives J = 37.2 meV and IPR 23.1. The criterion $J > \lambda/2$ for polaron hopping dominance [4] is *not* met at PBE+D3 ($J < \lambda/2$), so a polaron transition is not confirmed on this grid.

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[2] M. A. Khan, M. Atif, and M. N. Leuenberger, Tuning the electronic and optical properties of impurity-engineered two-dimensional graphullerene half-semiconductors, Phys. Rev. Mater. **9**, 034001 (2025).

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[5] R. M. Tromer, L. A. Ribeiro, Jr., and D. S. Galvão, A DFT study of the electronic, optical, and mechanical properties of a recently synthesized monolayer fullerene network, Chem. Phys. Lett. **804**, 139925 (2022).

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TABLE II. Positioning of the strain–dopant coupling energy $\mathcal{S}$ (main-text four-corner estimator) against common stability and electronic descriptors. Algebraically $\mathcal{S}$ has the form of an interaction (coupling) energy; the distinction is *dimensional*, i.e. which axes of the joint (ϵ, δ) control space each quantity resolves. "Additive-screening error" denotes the bias incurred when total energies under joint load are assembled additively from separate strain-only and doping-only scans; $\mathcal{S}$ equals that bias by construction. "Double-reference cross term" denotes a symmetric difference of differences over the four (ϵ, δ) corners rather than subtraction of a single reference state or fragment pair. Entries are conceptual scope, not numerical performance; consistent with its energetic definition, $|\mathcal{S}|$ is *not* reducible to a single geometric $(\Delta\bar{d})$ or charge (Δq) channel alone.

Descriptor	Nonlinear (ϵ, δ) coupling	Additive-screening error	Double-reference cross term	Joint strain + doping design
Formation / substitution energy E_{sub}	×	×	×	×
Binding / interaction energy ΔE_{int}	Partial	×	×	Partial
Linear strain coefficient α	×	×	×	Partial
Band gap E_g / work function	×	×	×	×
$\mathcal{S}$ (this work)	✓	✓	✓	✓

TABLE III. Archived legacy-PBE (no D3) tetramer cluster map retained for reproducibility only; not used for main-text quantitative claims. Cartesian coordinates are not affinely scaled. $E_{\mathrm{sub}} = (E_{\mathrm{doped}} - E_{\mathrm{pristine}})/n_{\mathrm{dopants}}$ at ϵ=0 (no chemical potentials; relative ranking within this protocol only).

System	n_{dop}	$E(\epsilon{=}0)$ (E_{h})	$E(\epsilon{=}+5\%)$ (E_{h})	α (meV/%)	E_{sub} (eV/dopant)
Pristine	—	−1363.5512	−1363.5509	−1.0	—
B-doped	13	−1328.4900	−1328.4090	+56.9	+73.4
N-doped	12	−1414.3812	−1414.4053	−303.2	−115.3
P-doped	12	−1370.4965	−1370.4888	−19.1	−15.8

Materials **7**, 3456 (2024).

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[10] P. Santra, S. Ghaderzadeh, M. Ghorbani-Asl, H.-P. Komsa, E. Besley, and A. V. Krasheninnikov, Strain-modulated defect engineering of two-dimensional materials, npj 2D Mater. Appl. **8**, 33 (2024).

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TABLE IV. DFT calculation groups supporting the main-text claims (reproducibility inventory; Supplemental Material only). Superseded exploratory grids appear in Table V.

Calculation group	System	Count	Protocol
Archived baseline (legacy PBE)	$4\times C_{60}$	24	Rigid strain; not used for main-text claims
π-DOS and gaps	$4\times C_{60}$	12	PBE+D3
IPR (polaron dimer)	Dimer	4	Pristine/B; ϵ=0, +3%
Hopping J	Dimer	4	PBE+D3
Charged polarons[a]	$4\times C_{60}$	20	UKS; geometry opt. + vertical SP
Periodic size scaling	$n\times C_{60}$	41	PBE+D3; 400/350 Ry
Cutoff benchmark	$6\times C_{60}$ N	1	+3%; 400 Ry single-point
Ionic relaxation	$4\times C_{60}$	4	Fixed-cell GEO; P corners
Ionic relaxation	$1\times C_{60}$ P	4	Fixed-cell GEO; periodic
Alternate placement	$4\times C_{60}$	24	Rigid PBE+D3
Periodic placement	$4\times C_{60}$	16	n=4; two independent maps
Matched-functional rigid	$4\times C_{60}$	4	PBE+D3 at relaxed coordinates
Reference placement	$4\times C_{60}$	24	Rigid PBE+D3
Hirshfeld population	$1\times C_{60}$ B/N/P	18	Spin-polarized KS

[a]Six relaxed tetramer geometries provide charged-state initial structures.

TABLE V. Exploratory DFT grids superseded by the production protocol (reproducibility archive only).

Task group	System	N_{conv}	Disposition
Early strain probe	$2\times C_{60}$	5	Superseded
Single-molecule doping	C_{60}	11	Superseded
Early electronics	$2\times C_{60}$	24	Superseded
Exploratory design grid	$4\times C_{60}$	20	Archive only

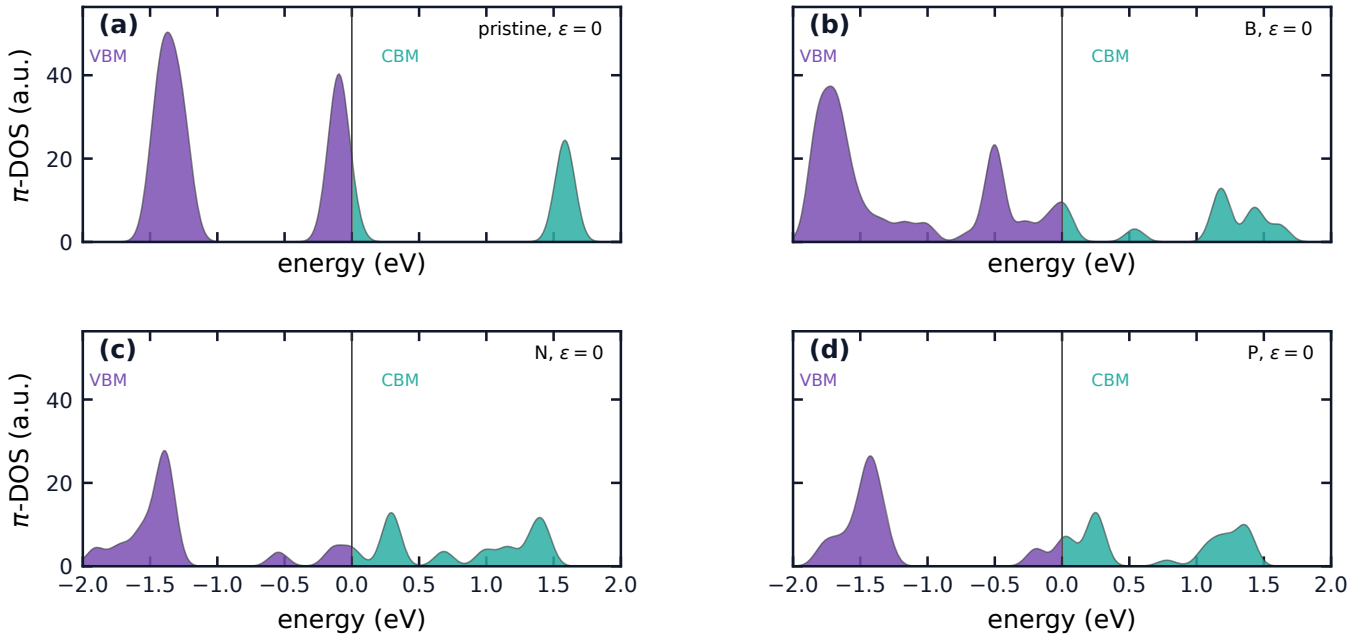
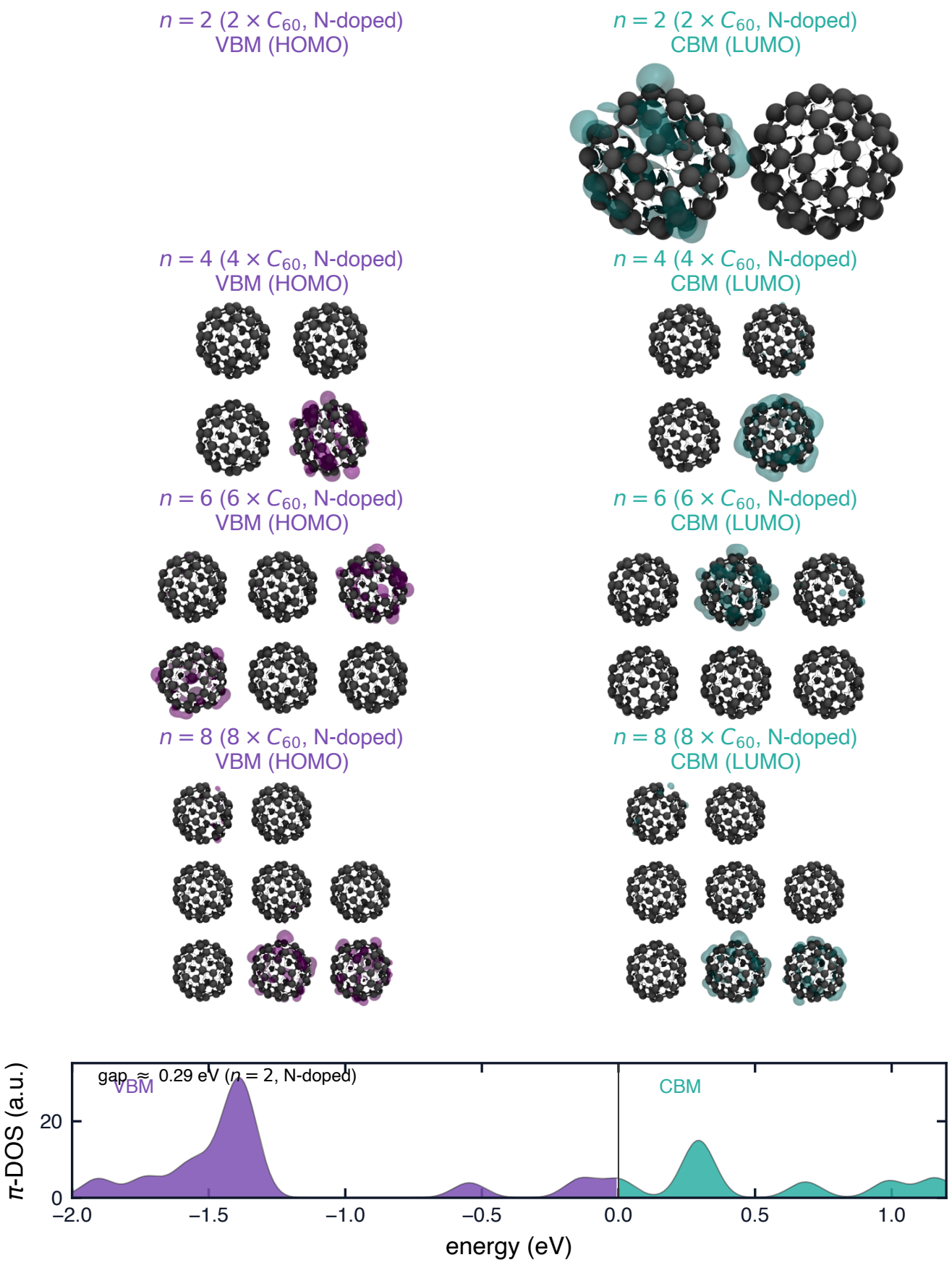


FIG. 1. Full π-DOS panels for pristine and B/N/P-doped tetramers at ϵ=0 (PBE+D3), complementary to the main-text electronic-structure figure.

qHP C_{60} graphullerene (ε = 0%, N-doped)



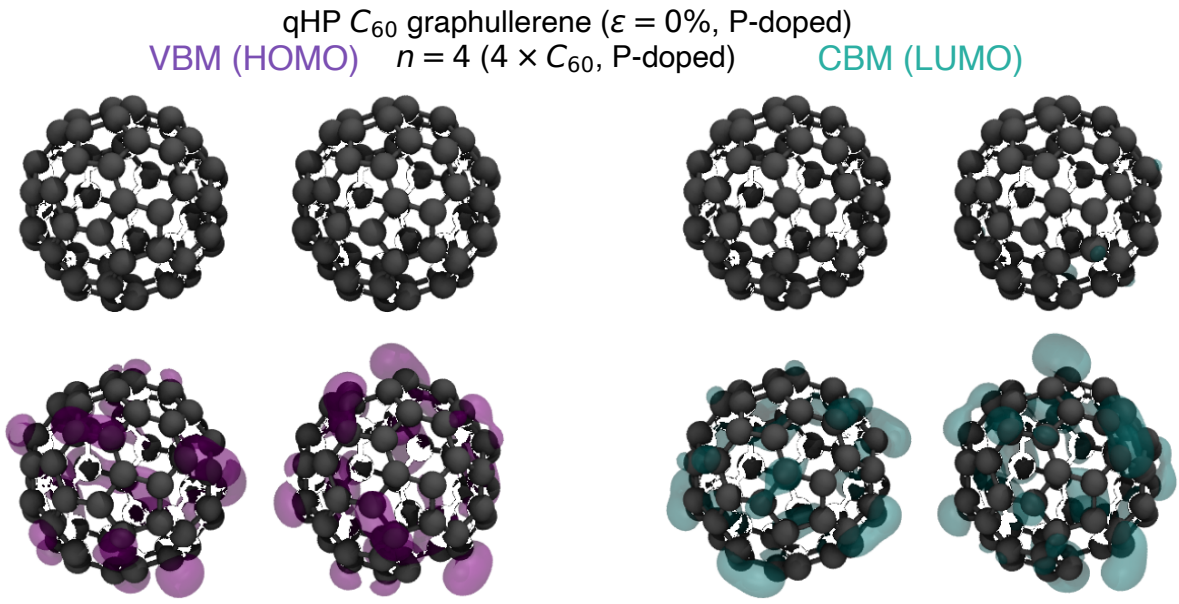


FIG. 3. P-doped n=4 supercell at ϵ=0: VBM (left) and CBM (right) on gray MSMS cages. Both frontier states localize on a subset of cages, providing a spatial counterpart to the large periodic $|\mathcal{S}|$ of size-mismatched P reported in the main text.

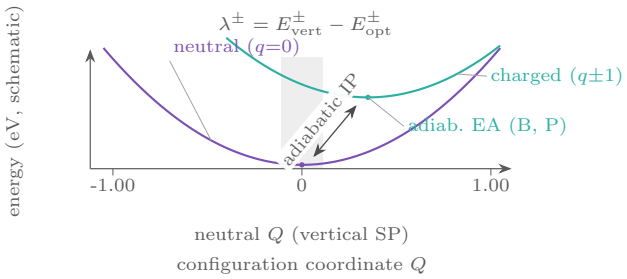
TABLE VI. Supplemental periodic coupling-energy rows at +3% ($n \geq 6$; 350 Ry production cutoff; completeness only).

n	Dopant	ϵ (%)	$\mathcal{S}$ (meV/atom)
6	B	+3	−7.76
6	N	+3	+2.38
6	P	+3	−31.89
8	B	+3	−11.26
8	N	+3	−2.30
8	P	+3	−29.94

TABLE VII. Machine-readable four-corner total energies per atom (Ha/atom) underlying $\mathcal{S}$ at +3% biaxial strain (PBE+D3, rigid protocol, reference placement seed 42). Corners follow Eq. (3): $E_{00} = E(0,0)$, $E_{\epsilon 0} = E(\Delta\epsilon, 0)$, $E_{0\delta} = E(0, \Delta\delta)$, $E_{\epsilon\delta} = E(\Delta\epsilon, \Delta\delta)$, all per atom relative to the same supercell size $n \times \mathrm{C}_{60}$. $\mathcal{S}$ (meV/atom) matches Eq. (2) and the open audit chain (CSV mirror in the public data release).

n	δ	ϵ (%)	E_{00}	$E_{\epsilon 0}$	$E_{0\delta}$	$E_{\epsilon\delta}$	$\mathcal{S}$
1	B	+3	-5.6826713370	-5.6825614149	-5.6339765778	-5.6341744697	−8.38
1	N	+3	-5.6826713370	-5.6825614149	-5.7532834847	-5.7530890722	+2.30
1	P	+3	-5.6826713370	-5.6825614149	-5.6922302079	-5.6932928189	−31.91
2	B	+3	-5.6817028549	-5.6817462745	-5.6338466634	-5.6339702803	−2.18
2	N	+3	-5.6817028549	-5.6817462745	-5.7532007583	-5.7529924491	+6.85
2	P	+3	-5.6817028549	-5.6817462745	-5.6895002016	-5.6905831259	−28.29
4	B	+3	-5.6828203886	-5.6827081746	-5.6341906516	-5.6343073761	−6.23
4	N	+3	-5.6828203886	-5.6827081746	-5.7535095848	-5.7532889091	+2.95
4	P	+3	-5.6828203886	-5.6827081746	-5.6924497414	-5.6932089910	−23.71
6	B	+3	-5.6828517956	-5.6827379495	-5.6340195050	-5.6341908867	−7.76
6	N	+3	-5.6828517956	-5.6827379495	-5.7533481235	-5.7531468231	+2.38
6	P	+3	-5.6828517956	-5.6827379495	-5.6916511984	-5.6927091482	−31.89
8	B	+3	-5.6828645675	-5.6825524079	-5.6341364256	-5.6342380770	−11.26
8	N	+3	-5.6828645675	-5.6825524079	-5.7534470201	-5.7532194756	−2.30
8	P	+3	-5.6828645675	-5.6825524079	-5.6919097127	-5.6926978180	−29.94

(a) Marcus protocol



(b) Adiabatic IP/EA and $\lambda^{\pm}$

Dopant	IP (eV)	EA (eV)	λ^{+} (eV)	λ^{-} (eV)
Pristine	4.73	2.61	0.041	0.042
N	3.89	2.77	0.058	0.065
B	4.11	3.12	0.051	—[†]
P	4.07	2.96	0.200	0.315

[†]B channel: $\lambda^{-} < 0$ under fixed neutral-geometry vertical SP; not reported.

FIG. 4. Marcus reorganization energies from charged-state geometry optimizations and vertical single-points at the neutral geometry. Panel (a) schematically defines the vertical–adiabatic decomposition following Ref. 4; panel (b) tabulates adiabatic IP/EA and $\lambda^{\pm} = E^{\pm}_{\mathrm{vert}} - E^{\pm}_{\mathrm{opt}}$ for pristine and B/N/P tetramers. The electron-channel λ^{-} for B is unphysical under the fixed neutral-geometry vertical protocol and is excluded (Limitations).

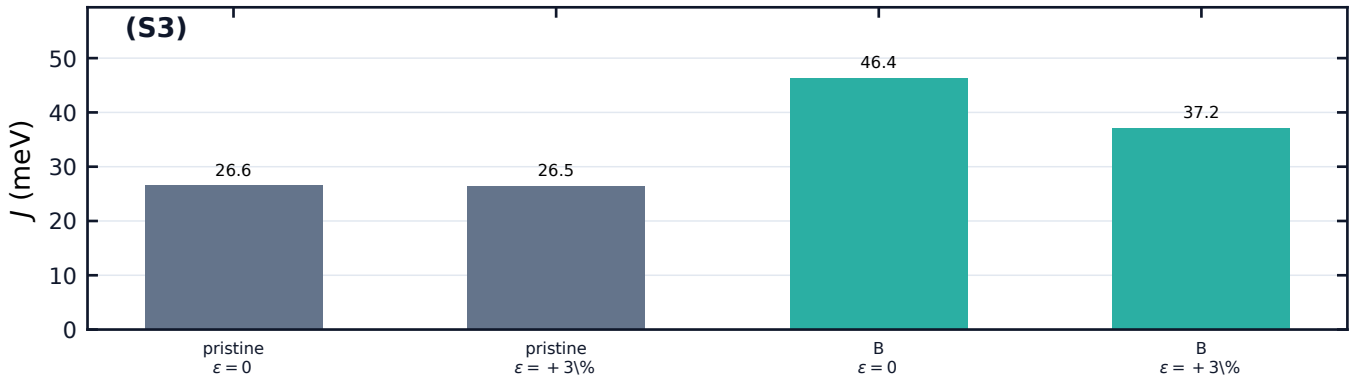


FIG. 5. Nearest-neighbor electronic coupling J on the tetramer dimer grid (pristine and B-doped cells at ϵ=0 and +3%; PBE+D3). Companion inverse-participation-ratio values for the same four corners are tabulated in the open data release (Methods).